

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 11 MATHEMATICS EXTENSION 1

COMMON TEST

MAY 2002

TIME ALLOWED: 70 MINUTES

Instructions:

- Show all necessary working
- Start each question on a new page
- Marks may be deducted for careless or badly arranged work.

Name: _____

Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total
/11	/10	/10	/11	/11	/9	/62

Question 1	Marks
(i) $(2 + 3\sqrt{3})(6 - 5\sqrt{3}) = a + \sqrt{b}$. Evaluate a and b.	2
(ii) Fully factorise $2a(b - 2c) - 3d(2c - b)$.	2
(iii) Factorise fully $20 - 9x - 20x^2$.	2
(iv) Simplify the fraction $\frac{1 - x^{-1}}{x^{-1} - x^{-2}}$	2
(v) Graph $y = x + 1$ and $y = x - 1 $ on the same number plane. Use this to solve the equation $ x + 1 = x - 1 $.	3

Question 2

- (i) (a) Show that $\frac{2x+1}{x+1}$ can be written as $2 - \frac{1}{x+1}$. 2
- (b) Hence, or otherwise find the equations of all asymptotes for the curve $y = \frac{2x+1}{x+1}$ 2
- (c) Sketch the curve showing intercepts and asymptotes. 2
- (ii) Calculate the cosine of the smallest angle of the triangle whose sides are 4cm, 5cm and 6cm. 2
- (iii) (a) Determine algebraically whether the function $f(x) = \frac{x}{1+x^2}$ is odd, even or neither. 1
- (b) If $f(x) = \begin{cases} -1 & x \leq -1 \\ x^2 & -1 < x < 1 \\ 1 & x \geq 1 \end{cases}$
- Evaluate $f(-1) + f(0) + f(1)$. 1

Question 3

- (i) (a) Find the domain of $f(x) = \frac{x^2 + 6x + 9}{x + 3}$ and by simplifying, find the range 2
- (b) Hence sketch the function. 2
- (ii) Solve $\frac{x}{x+3} \leq x$ 4
- (iii) Graph the region defined by $y \geq -\sqrt{16 - x^2}$ on the number plane. 2

Question 4

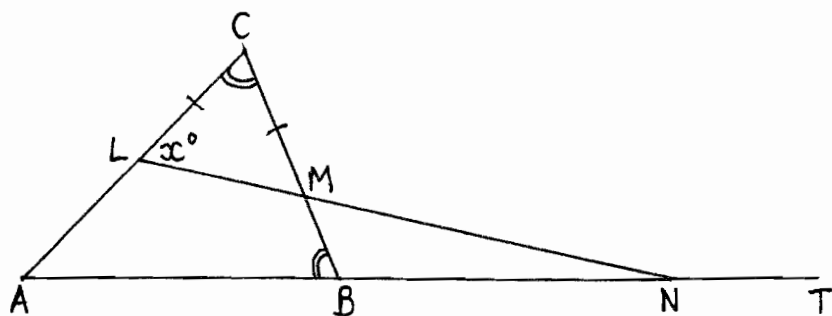
(i) Sketch the main features of;

(a) $y = (x-1)^3$ 2

(b) $3x - 2y - 4 = 0$ 2

(ii) Given $2^x = 8^{y+1}$ and $9^y = 3^x$, find x and y . 3

(iii) Given $CL = CM$, $\angle ACB = \angle ABC$ and $\angle CLM = x^\circ$,

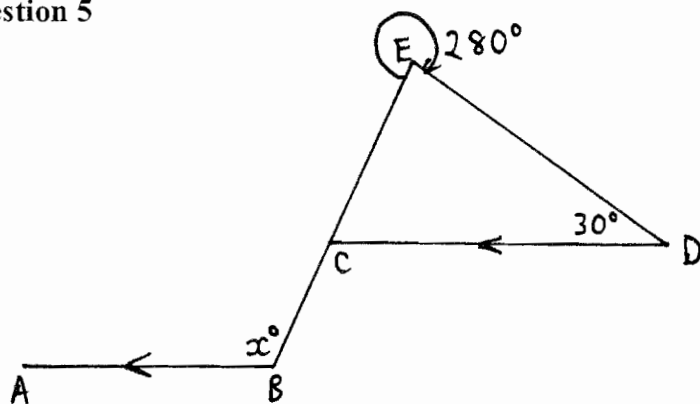


(a) show that $\angle ABC = 180 - 2x$ 2

(b) Hence show $\angle TNL = 3x^\circ$ 2

Question 5

(i) Find x giving reasons 2



(ii) Sketch $y = \frac{|x|}{x^2}$ 2

(iii) Simplify $\frac{\sin(180 + \theta)}{\sin(90 - \theta)}$ 2

(iv) Find the exact value of $\cos(-150^\circ)$

Q5 continued

(v) Two yachts sail in a straight line from a bouy B. The first sails 10km in the direction 040°T , and the second sails 20km in the direction 140°T in the same time.

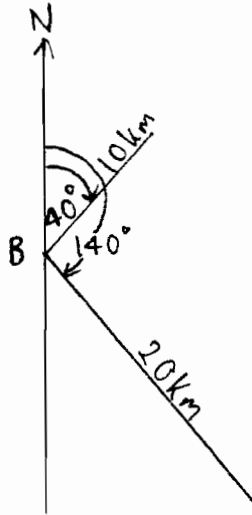
(a) Copy the diagram below onto your answer sheet.

(b) How far apart are they? (correct to nearest km)

2

(c) What is the bearing of the 1st yacht from the 2nd? (correct to nearest degree)

2



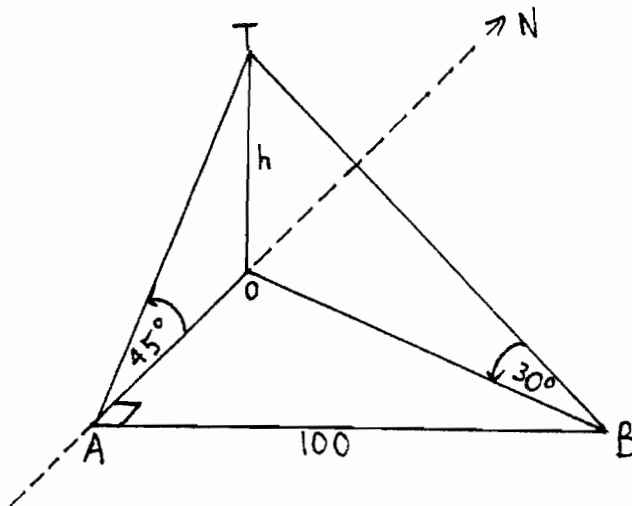
Not to scale

Question 6

(i) In the triangle PQR, $PQ = 18\text{cm}$, $QR = 24\text{cm}$ and $\angle R = 12^\circ$. Find the possible size(s) of $\angle P$ correct to the nearest degree.

3

(ii)



Not to scale

A surveyor stands at a point A, which is due south of a Tower OT of height h m. The angle of elevation of the top of the tower from A is 45° . The surveyor then walks 100m due east to point B, from where she measures the angle of elevation of the top of the tower to be 30° .

(a) Express the length OB in terms of h .

1

(b) Show that $h = 50\sqrt{2}$.

3

(c) Calculate the bearing of B (to nearest degree) from the base of the tower.

2

Solution to Extension Task 1 (Preliminary) 2002

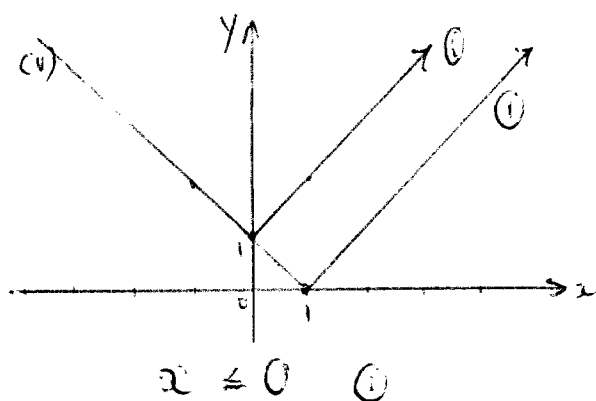
Question 1

$$\begin{aligned} \text{a)} \quad & (2+3\sqrt{3})(6-5\sqrt{3}) \\ & = 12 - 10\sqrt{3} + 18\sqrt{3} - 15 \times 3 \\ & = -33 + 8\sqrt{3} \quad \textcircled{1} \\ & = -33 + \sqrt{192} \\ \therefore \quad & a = -33, \quad b = 192 \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{a)} \quad & 2a(b-2c) - 3d(2c-b) \\ & 2a(b-2c) + 3d(b-2c) \quad \textcircled{1} \\ & (2a+3d)(b-2c) \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{a)} \quad & 20 - 9x - 20x^2 \\ & -20x^2 - 9x + 20 \\ + \quad & \begin{array}{r} -9 \\ -400 \\ -25, 16 \end{array} \quad \begin{array}{l} (-20x-25)(-20x+16) \\ \hline -20 \\ -25(4x+5) + (4-5x) \\ \hline -20 \end{array} \\ & (4x+5)(4-5x) \quad \textcircled{1} \end{aligned}$$

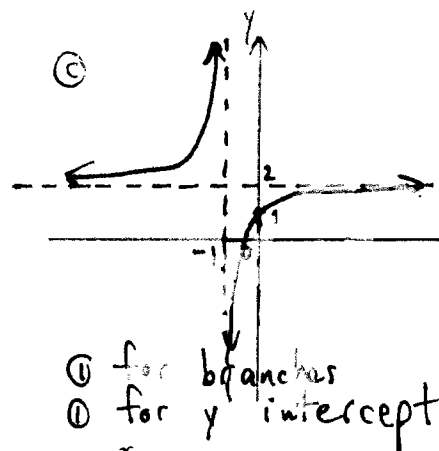
$$\begin{aligned} \text{a)} \quad & \frac{1-x^{-1}}{x^{-1}-x^{-2}} \\ & = \frac{1-\frac{1}{x}}{\frac{1}{x}-\frac{1}{x^2}} \times \frac{x^2}{x^2} \\ & = \frac{x^2-x}{x-1} \quad \textcircled{1} \\ & = \frac{x(x-1)}{(x-1)} \\ & = x \quad \textcircled{1} \end{aligned}$$



Question 2

$$\begin{aligned} \text{a)} \quad & \textcircled{a} \quad \frac{2x+1}{x+1} \\ & = \frac{2(x+1)}{x+1} - \frac{1}{x+1} \quad \textcircled{1} \\ & = 2 - \frac{1}{x+1} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad & x = -1 \quad \textcircled{1} \\ & y = 2 \quad \textcircled{1} \end{aligned}$$



$$\begin{aligned} \text{a)} \quad & \cos \theta = \frac{6^2 + 5^2 - 4^2}{2 \times 6 \times 5} \quad \textcircled{1} \\ & \cos \theta = \frac{3}{4} \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{a)} \quad & f(x) = \frac{x}{1+x^2} \\ & \text{odd if } f(-x) = -f(x) \\ & -x \quad -x \quad \textcircled{1} \end{aligned}$$

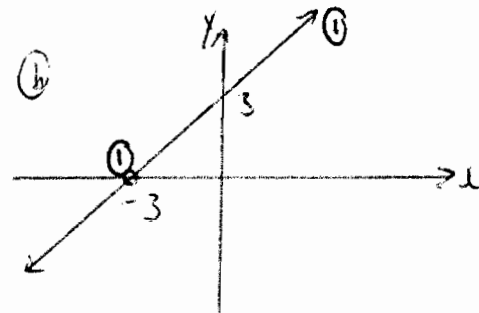
$$\begin{aligned} \textcircled{b} \quad & f(-1) + f(0) + f(1) \\ &= -1 + 0^2 + 1 \\ &= 0 \quad \textcircled{1} \end{aligned}$$

Question 3

$$\textcircled{a} \quad f(x) = \frac{x^2 + 6x + 9}{x+3}$$

D: All real x , $x \neq -3$ $\textcircled{1}$

R: All real y , $y \neq 0$ $\textcircled{1}$



$\textcircled{ii} \quad \frac{x}{x+3} \leq x$
Critical pts. $x = -3$ $\textcircled{1}$

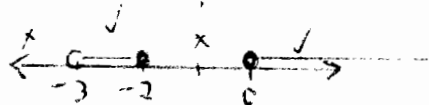
$$\frac{x}{x+3} = x$$

$$x = x^2 + 3x$$

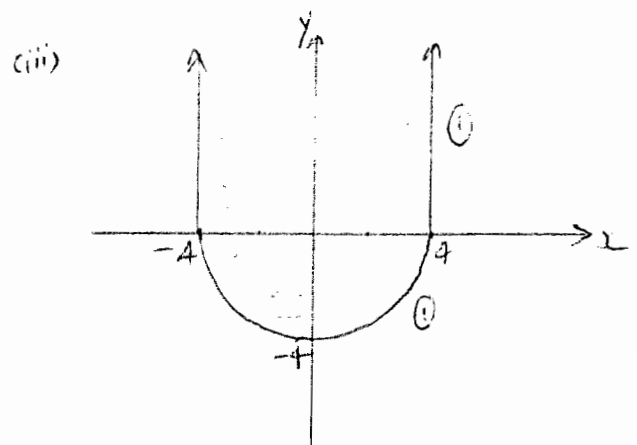
$$0 = x^2 + 2x$$

$$0 = x(x+2)$$

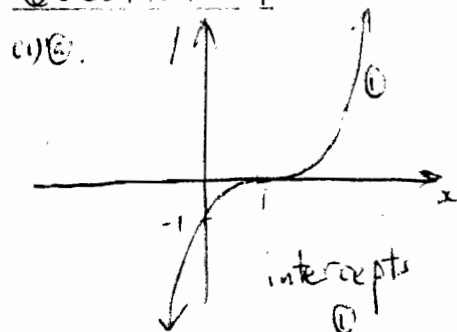
$$x = 0, -2 \quad \textcircled{2}$$



$$x \geq 0, -3 < x \leq -2 \quad \textcircled{1}$$

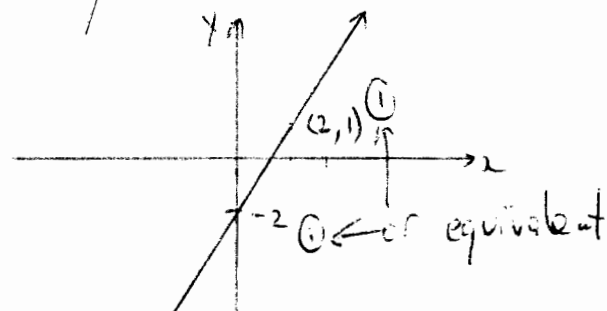


Question 4



$$\textcircled{b} \quad 3x - 2y - 4 = 0$$

$$y = \frac{3}{2}x - 2$$



$$\begin{aligned}
 \text{cii) } 2^x &= 8^{y+1} & 9^y &= 3^x \\
 2^x &= 2^{3(y+1)} & 3^{2y} &= 3^x \quad \textcircled{1} \\
 \Rightarrow x &= 3y+3, \\
 x &= 2y \\
 \Rightarrow 3y+3 &= 2y \quad \textcircled{1} \\
 y &= -3 \therefore x = -6 \quad \textcircled{1}
 \end{aligned}$$

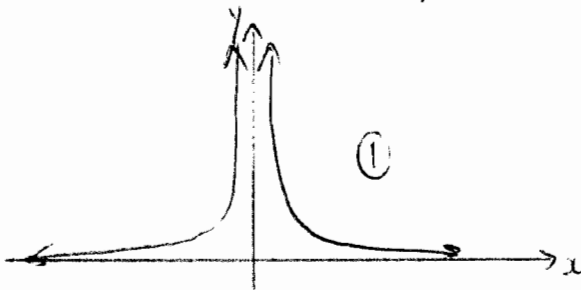
$$\begin{aligned}
 \text{cii) } \angle LMC &= x^\circ \text{ (base } \angle \text{'s of isos. } \triangle \text{)} \quad \textcircled{1} \\
 \angle LCM &= 180 - 2x \text{ (}\angle \Sigma \text{ of } \triangle \text{)} \\
 \angle ABC &= 180 - 2x \text{ (base } \angle \text{'s of isos. } \triangle ABC \text{)} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \angle BMN &= x \text{ (vert. opp } \angle \text{'s)} \\
 \angle MBN &= 2x \text{ (}\angle \Sigma \text{ on straight line)} \quad \textcircled{1} \\
 \angle TNL &= 2x + x \\
 &= 3x \text{ (Ext. } \angle \text{ of } \triangle \text{)} \quad \textcircled{1}
 \end{aligned}$$

Question 5

$$\begin{aligned}
 \text{ci) } \angle BCD &= x \text{ (alt. } \angle \text{'s = in } \parallel \text{ lines)} \\
 \angle ECA &= 180 - x \text{ (}\angle \Sigma \text{ on straight line)} \\
 \angle CED &= 180 - 30 - (180 - x) \\
 &= x - 30 \quad \textcircled{1} \\
 &= 80 \text{ (revolution = } 360 \text{)} \\
 x &= 110^\circ \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{cii) } y &= \frac{|x|}{x^2} = \frac{1}{x}, \quad x > 0 \quad \textcircled{1} \\
 &= -\frac{1}{x}, \quad x < 0
 \end{aligned}$$



$$\begin{aligned}
 \text{cii) } \frac{\sin(180 + \theta)}{\sin(90 - \theta)} \\
 &= \frac{-\sin \theta}{\cos \theta} \quad \textcircled{1} \\
 &= -\tan \theta \quad \textcircled{1}
 \end{aligned}$$

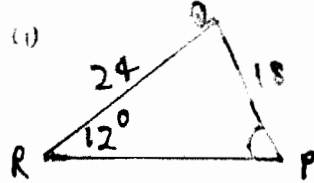
$$\begin{aligned}
 \text{(v) } c^2 &= a^2 + b^2 - 2ab \cos C \\
 c^2 &= 10^2 + 20^2 - 2 \cdot 10 \cdot 20 \cdot \cos 100^\circ \quad \textcircled{1} \\
 c^2 &= 569.4 \\
 c &= 24 \text{ km} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{civ) } \cos(-150^\circ) \\
 &= \cos 210^\circ \\
 &= -\cos 30^\circ \\
 &= -\frac{\sqrt{3}}{2} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } \frac{\sin \theta}{10} &= \frac{\sin 60^\circ}{24} \\
 \theta &= 24^\circ \quad \textcircled{1}
 \end{aligned}$$

Question 6

(1)



$$\frac{\sin P}{24} = \frac{\sin 12}{18} \quad (1)$$

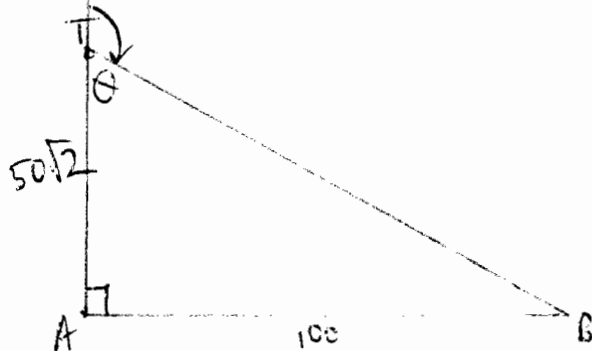
$$P = 16^\circ \text{ or } 164^\circ$$

(1) (1)

$$\begin{aligned} \text{iii) } \tan 30 &= \frac{h}{OB} \\ \frac{1}{\sqrt{3}} &= \frac{h}{OB} \\ OB &= h\sqrt{3} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{(2) Similarly } \tan 45 &= \frac{h}{OA} \\ OA &= h \quad (1) \\ \text{By Pythagoras'} \\ OB^2 &= OA^2 + 100^2 \\ h^2 3 &= h^2 + 10000 \quad (1) \\ 2h^2 &= 10000 \\ h^2 &= \frac{5000}{2} \\ h &= \sqrt{5000} \\ h &= 50\sqrt{2} \text{ as req'd. } (1) \end{aligned}$$

(c) Top View



$$\begin{aligned} \tan \theta &= \frac{100}{50\sqrt{2}} \\ \theta &= 54^\circ 44' \quad (1) \end{aligned}$$

$$\begin{aligned} \therefore \text{Bearing} &= 180 - 54^\circ 44' \\ &= 125^\circ \text{ from true north. } (1) \end{aligned}$$